

Comptia A+ Assignment

[A+ - Troubleshooting and Helpdesk]

Component-Specific Troubleshooting

Identify and fix issues in faulty components (RAM, GPU, storage). Create a checklist for diagnosing hardware issues.

Task

- ❖ **1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.**
 - To safely remove and reinstall (reseat) the RAM module and verify that the computer recognizes the memory correctly after reassembly.
 - Tools Required:
 - Screwdriver (if required)
 - Soft cloth or clean workspace
 - Step 1: Shut Down the Computer:
 - Turn off the computer completely, unplug the power cable, and remove the laptop battery (if removable).



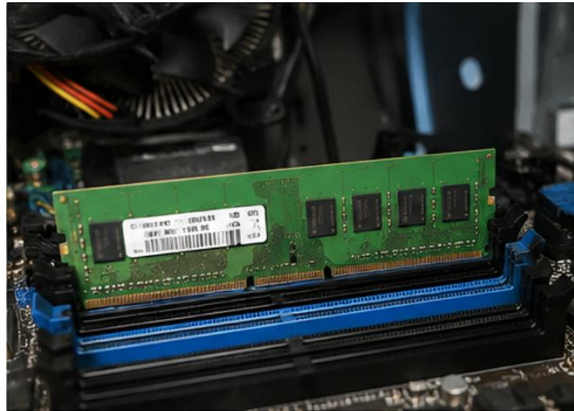
➤ Step 2: Open the Computer Case:

- Carefully remove the side panel of the desktop (or the RAM access cover on a laptop) to access the motherboard.



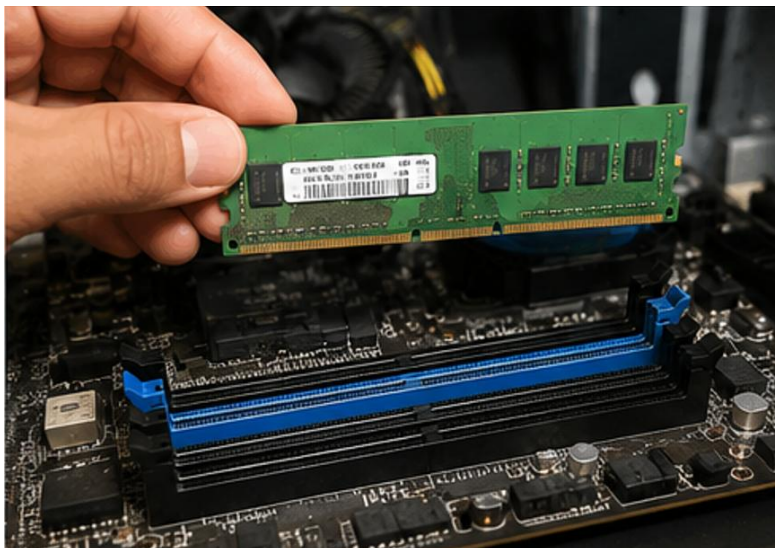
➤ Step 3: Locate the RAM Module:

- Identify the installed RAM module and the retaining clips on both sides.



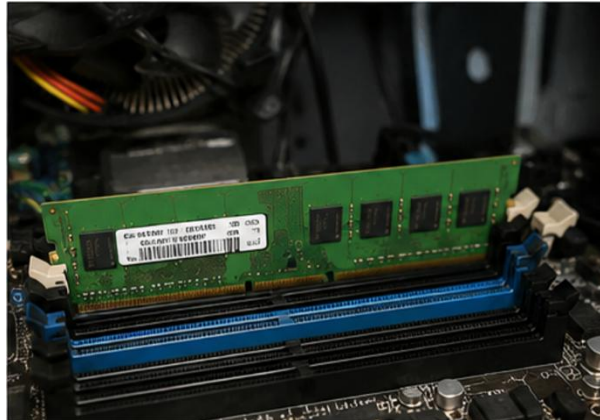
➤ Step 4: Remove the RAM Module:

- Press the retaining clips outward. The RAM module will pop up slightly. Hold it by the edges and remove it carefully.



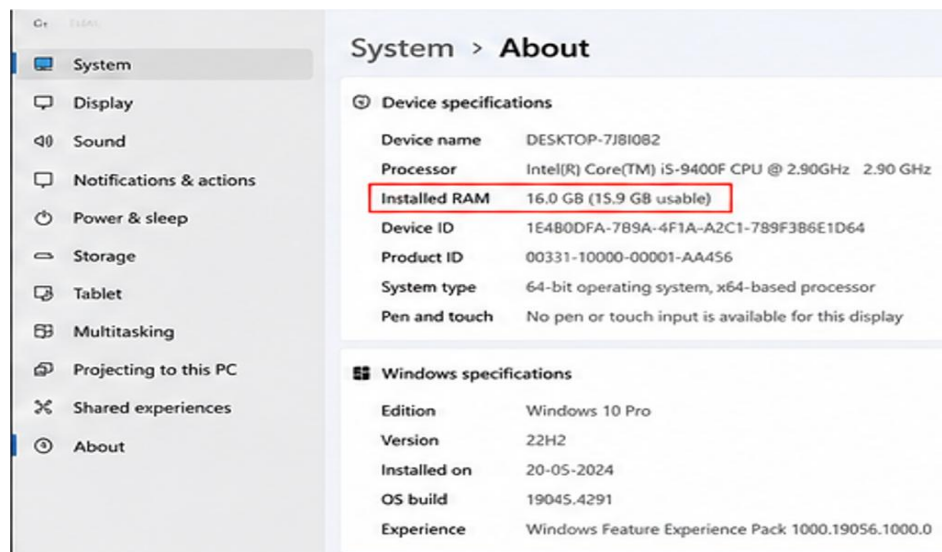
➤ Step 5: Reinsert the RAM Module:

- Align the notch on the RAM module with the slot and press down evenly until both retaining clips lock into place.



➤ Step 6: Close the Computer and Test:

- Reassemble the computer, reconnect the power cable, and turn the system on. Verify that the computer starts normally and detects the installed RAM.



- ❖ **2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Think about display issues, driver checks, and physical connections.**

➤ GPU Troubleshooting Checklist:

Checklist	Step	What to Check / Action
✓	1. Identify the Symptoms	Check for <u>no display</u> , <u>screen flickering</u> , <u>graphical artifacts</u> , <u>crashes</u> , or <u>low gaming performance</u> . Note when the problem occurs.
✓	2. Check the Monitor and Display Cable	Make sure the monitor is powered on and the <u>HDMI/DisplayPort cable</u> is firmly connected to the GPU and monitor. If possible, test another cable or display port.
✓	3. Check the GPU Physical Connection	Turn off and unplug the PC. Open the case and make sure the GPU is properly seated in the <u>PCIe slot</u> and its PCIe power connectors are firmly connected.
✓	4. Check the GPU Drivers	In Windows, open <u>Device Manager</u> → <u>Display</u>

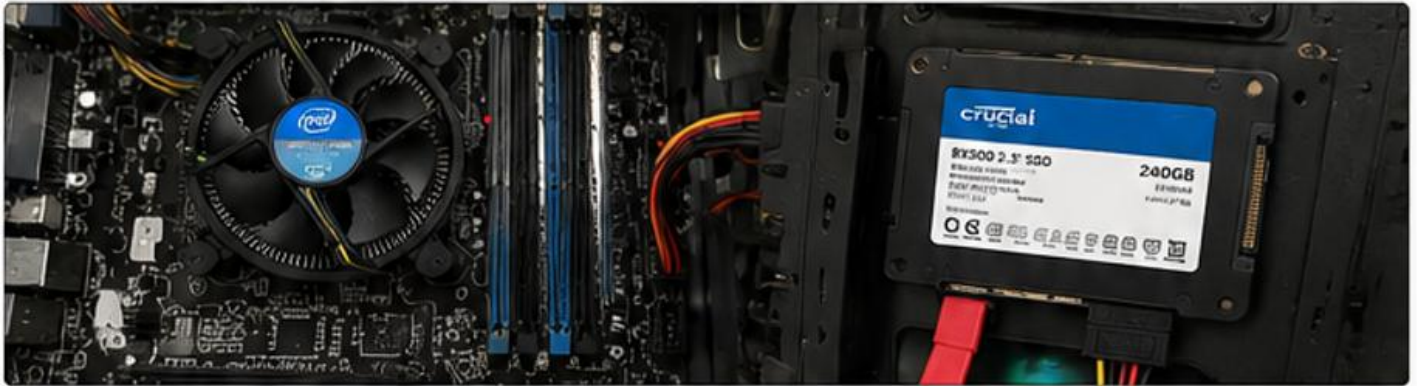
		<u>adapters</u> and check whether the GPU is detected. Look for warning/error icons and update or reinstall the graphics driver if necessary.
✓	5. Check GPU Cooling	Check whether the GPU fans are spinning and whether there is excessive dust blocking airflow. Overheating can cause crashes, shutdowns, or graphical problems.
✓	6. Test the GPU	Run a graphics-intensive game or a suitable GPU diagnostic/stress test and observe whether the display remains stable without artifacts or crashes.
✓	7. Confirm the Fix	Restart the PC and verify that the GPU is detected correctly, the display works normally, and the game runs without crashes, flickering, or artifacts.

❖ **3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**

- To simulate a storage connectivity problem by disconnecting a SATA SSD and then reconnecting it to observe the BIOS and startup behavior.
- Step 1: Shut Down the Computer:
 - First, shut down the computer completely and switch off the power supply. Disconnect the power cable before opening the cabinet.



- Step 2: Open the Computer Cabinet:
 - Remove the side panel carefully to access the internal components and locate the SATA SSD.



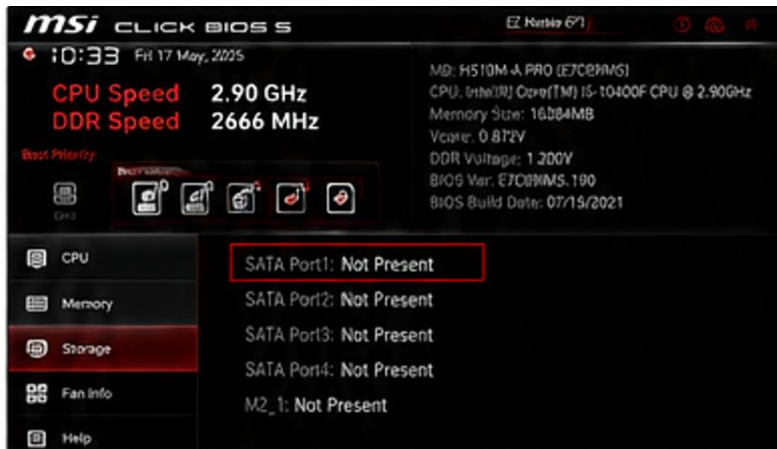
➤ Step 3: Disconnect the SATA SSD:

- Locate the SATA data cable and SATA power cable connected to the SSD. Disconnect both cables carefully to simulate a storage connectivity failure.



➤ Step 4: Power On and Check the Error:

- Close the cabinet and power on the computer. Enter the BIOS/UEFI setup and check whether the SSD is detected.



BIOS Screen (Before Reconnection)

Observation: In BIOS, SATA Port1 shows “Not Present” which indicates the SSD is not detected.



Startup Error Message (Before Reconnection)

Observation: System shows “A2: SATA Device Not Found” error during startup.

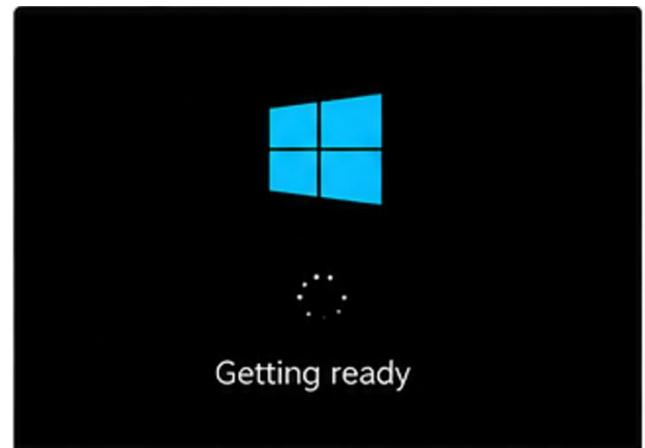
➤ Step 5: Reconnect the SATA SSD:

- Turn off the computer again. Reconnect both the SATA data cable and SATA power cable firmly to the SSD. Ensure that both connections are secure.



➤ Step 6: Power On and Verify the Result:

- Power on the computer and enter the BIOS/UEFI setup again. Check the SATA information.
- After the Reconnecting the SSD.



- The SSD is detected correctly.
- The drive name appears in the BIOS.
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❖ **4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.**

➤ Component Checked First: - **RAM**

➤ Reasoning:

- I would check the RAM first because faulty, loose, or poorly seated RAM can cause system instability, freezing, crashes, unexpected restarts, and sometimes POST errors. It is also relatively easy to inspect and reseat compared with other components.

➤ Step-by-Step RAM Diagnostic Procedure:

➤ Step 1: Shut Down and Disconnect the PC:

- Turn off the PC completely, switch off the power supply, and disconnect the power cable.

➤ What to Check:

- Ensure there is no power going to the system before opening the case.

➤ Step 2: Inspect the RAM Physically:

- Open the computer case and locate the RAM modules on the motherboard.

➤ What to Check:

- RAM is properly seated.
- Retaining clips are locked.
- No visible physical damage.

- RAM slot is free from excessive dust.
- Step 3: Reseat the RAM:
 - Release the side clips and carefully remove the RAM module. Reinsert it firmly until both retaining clips lock into place.
- What to Check:
 - Make sure the RAM is aligned correctly with the notch in the slot and is completely seated.
- Step 4: Test RAM Modules Individually:
 - If the PC has two or more RAM modules, test one module at a time.
- What to Check:
 - Boot the PC with only one RAM module installed.
 - If the restart problem continues, test the other module.
 - Try another compatible RAM slot if necessary.
 - This can help identify whether one particular RAM module or slot is faulty.
- Step 5: Check for POST Errors:
 - Power on the PC and observe the startup process.
- What to Check:
 - Beep codes
 - No-display condition
 - Memory-related POST error messages
 - Whether the PC boots normally

➤ Step 6: Confirm the Fix:

- After reseating/testing the RAM, use the computer normally and check whether the random restarts have stopped.

➤ Expected Result:

- If the PC becomes stable and no longer restarts unexpectedly, the RAM seating or RAM module may have been the cause.